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Editor-in-Chief
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Dear Editor,

I am pleased to submit the manuscript, “Fair, Non-Progressive Influence Control Under Backfire and Parameter Uncertainty,” for consideration for publication in IEEE Transactions on Knowledge and Data Engineering.

This paper studies influence maximization under a combined relaxation of the four assumptions underlying the classical Kempe–Kleinberg–Tardos guarantee: progressive diffusion, monotone submodularity, known influence parameters, and the absence of a fairness requirement. Real deployments routinely violate all four at once – influence can be actively reversed through negative word-of-mouth, edge probabilities are estimated rather than known, and social interventions must not neglect a disadvantaged subpopulation. We make the following contributions:

- A sandwich approximation built from two monotone-submodular surrogates, giving a $\rho(1 - 1/e)$ guarantee for every backfire intensity $\beta \in [0, 1)$, with a directly computable per-instance certificate, and a proof that $(1 - 1/e)$ -hardness survives backfire exactly.
- A closed-form Lagrangian index for the resulting 2-state/3-action per-node control problem, replacing a bisection-plus-value-iteration inner loop with an $O(Km + n \log n)$ -per-round procedure – roughly two orders of magnitude faster in our measurements.
- Identification and correction of a population-weighting pathology in an existing isoelastic-welfare fairness objective that systematically under-protects small groups.
- MF-BWI-Fair, a control policy combining these results with mean-field belief propagation and Bayesian parameter tracking, validated against independently reimplemented published baselines (including a direct benchmark of our IMM reimplementations against the original authors’ own C++ code) on a primary synthetic benchmark, four further synthetic graphs, and three real Facebook ego-networks at increasing scale.

We believe this work is well suited to the scope of IEEE Transactions on Knowledge and Data Engineering: it connects submodular/RIS-style influence-maximization theory to restless-bandit sequential control theory – a connection we believe is novel for a node-level activation/maintenance control problem – while remaining grounded in classical, guarantee-preserving algorithmic techniques and a thorough, honestly reported empirical evaluation.

This manuscript is original, has not been published previously, and is not currently under consideration for publication elsewhere. I have no conflicts of interest to declare. Thank you for considering

our submission; I look forward to the reviewers' feedback.

Sincerely,

Quang-Vinh Dang